we need to do a great deal of useful physics, since it fixes properties such as the temperature and pressure of the system. But the distribution isn't a complete description of the system; it's simply a reflection of what we know (or don't) about it. To tag this distinction with philosophical buzzwords, in statistical mechanics the probability distribution is an *epistemic* notion—describing the state of our knowledge—rather than an *ontological* one—describing some objective feature of reality. Epistemology is the study of knowledge; ontology is the study of what is real.

It was natural, in 1927, to suspect that quantum mechanics should be thought of along similar lines. After all, by that time we had figured out that what we use wave functions for is to calculate the probability of any particular measurement outcome. Surely it makes sense to imagine that nature itself knows precisely what the outcome is going to be, but the formalism of quantum theory simply doesn't completely capture that knowledge, and thus needs to be improved. The wave function, in this view, isn't the whole story; there are additional "hidden variables" that fix what the actual measurement outcomes are going to be, even if we don't know (and perhaps can't ever determine ahead of the measurement) what their values are.

Maybe. But in subsequent years a number of results have been obtained, most notably by the physicist John Bell in the 1960s, implying that the most simple and straightforward attempts along these lines are doomed to failure. People tried—de Broglie actually put forward a specific theory, which was rediscovered and extended by David Bohm in the 1950s, and Einstein and Schrödinger both batted around ideas. But Bell's theorem implies that any such theory requires "action at a distance"—a measurement at one location can instantly affect the state of the universe arbitrarily far away. This seems to be in violation of the spirit if not the letter of the theory of relativity, which says that objects and influences cannot propagate faster than the speed of light. The hidden-variable approach is still being actively pursued, but all known attempts along these lines are ungainly and hard to reconcile with modern theories such as the Standard Model of particle physics, not to mention speculative ideas about quantum gravity, as we'll discuss later. Perhaps this is why Einstein, the pioneer of relativity, never found a satisfactory theory of his own.